

Verifier Reach Is Spec Reach:

Moving Defect-Naming Forward in Agentic Coding Pipelines

David Sienkowski
research and measurements

Draft, 2026-06-26 — experience/position paper

Abstract

Large-language-model verifiers are increasingly trusted to judge AI-generated code, yet they systematically miss defects the specification never named—and do so confidently. On a corpus of *non-inferable* defects, where the correct behavior cannot be derived from the specification plus general knowledge, a held-out test oracle caught 100% (3/3 tasks) while a goal-backward LLM verifier caught 0 of 12 verdicts, and was severely miscalibrated *there* (ECE 0.81 on the clean non-inferable tasks versus 0.03 on inferable tasks). The blind spot is *model-invariant*: it reproduces across three model tiers and across several independent experiments, and a frontier model does no better than chance on it once a tie-break scoring artifact is removed. A powered replication ($n = 121$ verdicts, three model tiers) reproduces the blind spot at a 94% confident-false-pass rate and shows that resolving the surfaced edge into the spec converts it to a 94% catch; an independent, CI-committed 43-item eval built by the framework’s maintainer corroborates this and refines it—the fix is *exogenous* tagging, not endogenous self-disconfirmation (which can even regress a correct verdict). We argue the binding constraint is **specification reach**, not verifier reasoning: a verifier can only catch what the spec named. We present a productized response shipped into an open-source agentic coding framework (GSD-Core): (1) a shape-classified *edge-probe* that surfaces omitted edge cases at specification time; (2) an adversarial *prohibition-probe* that elicits must-NOT constraints into a structured, verification-tiered contract; and (3) fail-closed verification that hard-gates mechanically-checkable constraints and, for judgment-tier constraints it cannot confirm, abstains (`insufficient_spec`) rather than false-passes. We describe the unifying verification-tier abstraction, deployment via maintainer-reviewed merges, and—honestly—the residual class where the probe’s recall gap leaves a held-out backstop irreplaceable. The contribution is a novel synthesis plus original measurement, not a net-new concept; the results are direction-finding on small samples plus a reproducible harness, not powered studies.

1 Introduction

Agentic coding pipelines increasingly close the loop with an LLM *verifier*: after a model writes code, another model (or the same one in a fresh role) reads the code, the specification, and the test artifacts, and renders a verdict—pass, or gaps found. The appeal is obvious. A verifier needs no test harness, generalizes across languages, and reads intent the way a human reviewer does. The risk is just as obvious in hindsight: a verifier that *reasons* about correctness can be confidently wrong, and unlike a failing test it offers no ground truth to contradict it.

This paper makes one claim and builds a system around it:

A verifier’s reach is the specification’s reach. An automated verifier can only catch a defect whose correct behavior is recoverable from the spec plus general knowledge.

When the spec omits a domain edge, the verifier does not flag the omission—it fills the gap with a plausible guess, marks the code passing at high confidence, and the bug ships unflagged.

The lever this implies is counterintuitive. The instinct, when a verifier misses bugs, is to make the *judge* smarter: a stronger model, an ensemble, a debate, a calibrated confidence gate. We show that on the defects that actually slip through, every one of those downstream moves is a no-op, because the information the verifier needs was never in its input. The productive move is to widen the input: move defect-naming **forward**, to specification time, so that the obligation the verifier would have missed becomes something it can check.

Contributions.

1. **The measurement** (§2): on a purpose-built non-inferable corpus, a goal-backward verifier catches 0/12 while a held-out oracle catches 100%, and is decisively miscalibrated only in the non-inferable regime. The blind spot is model-invariant.
2. **The design principle** (§3): verifier reach = spec reach. Convert non-inferable obligations into inferable ones at spec time, rather than strengthening the judge.
3. **A shipped, tiered, fail-closed pipeline** (§4): an edge-probe and an adversarial prohibition-probe that write a verification-tiered `must_haves` contract, plus a verifier that abstains rather than false-passes on what it cannot confirm.
4. **Honest residual analysis** (§7): the probe’s recall is classifier- and model-tier-dependent; a held-out backstop remains necessary for the residue.

We frame these as *confirmations and direction-finding plus a reproducible harness*, not as discoveries. The samples are small (single- to double-digit per cell) and drawn from one framework; the effects are large and one-sided, but they are leads, not proofs.

2 The Measurement

Non-inferable corpus. We built three self-validating tasks whose correct behavior cannot be inferred from the spec we wrote plus general programming knowledge: half-to-even (banker’s rounding, touching-interval merge ($[1, 2]$ and $[2, 3]$ merge or not?), and grapheme-aware string truncation. A validation script proves each corpus is well-formed: the reference implementation passes a held-out test that encodes the omitted edge, while the defective implementation passes the visible spec-derived tests only. “Non-inferable” is relative to the spec given—a more complete spec would make these inferable, which is precisely the point.

T1: held-out catches what the verifier cannot. Twelve inspection-only verifier subagents (opus and haiku, each emitting a verdict and a calibrated confidence) reviewed the defective code. Table 1 states both denominators.

On the touching-interval task, all four verifiers passed the defective code, one explicitly reasoning that `strict <` is “defensible under a standard overlap definition”—a textbook non-inferable miss: the spec never defined *touching*, so there was no basis to flag it.

Table 1: Catch rate of the non-inferable defect.

Mechanism	caught the omitted edge
Held-out test (encodes the omitted edge)	3/3 tasks = 100% (deterministic)
Goal-backward verifier (spec + general knowledge)	0/12 verdicts

T2: the blind spot is a calibration failure, not a coverage gap. The same protocol scores the verifier’s calibration across regimes (Table 2). On inferable defects the verifier is essentially perfect and well-calibrated. On the non-inferable clean tasks it is right 13% of the time while 93% confident—an expected calibration error of 0.81, and a Brier score worse than chance. The verifier expresses ≈ 0.93 confidence whether it is right (inferable) or wrong (non-inferable): **confidence is uninformative across regimes**. A confidence gate cannot help, because the verifier is just as sure when it is blind.

Table 2: Verifier calibration by regime (defective code; ground truth = gap exists).

Regime	n	accuracy	mean conf.	ECE	Brier
Inferable defects	16	100%	0.970	0.030	0.001
Non-inferable (all tasks)	12	25%	0.927	0.677	0.644
Non-inferable (clean 02–03)	8	13%	0.930	0.805	0.753

The blind spot is model-invariant. The headline corpus is small, but the blind spot recurs across independent experiments, each attacking the claim from a different angle:

- A verifier-reach gauge (36 inspection verdicts across three tiers) finds true reach—verdict *and* correct identification of the planted edge—at 89% on inferable defects but 0% on genuinely-arbitrary edges, unanimous across tiers.
- A model-version probe finds uniform blindness across opus/sonnet/haiku (the omitted edge caught 1/9); verdicts barely drift, but confidence-while-wrong *rises as model strength falls* (opus ≈ 0.65 versus haiku ≈ 0.96).
- An ordering-resolvability study finds non-inferable recovery at or barely above chance even at the frontier (14–29% against a 16.7% floor, the small excess a tie-break scoring artifact); a $\sim 30\times$ cost increase buys no additional recovery.
- A repeatable-execution harness finds opus $\equiv$ haiku on an omitted rounding tie (capability 75%, reliability 0%, byte-identical across reruns).

Re-running a bigger model never closes the gap; only spec reach does.

A powered replication (n=121). We replicate the blind spot at scale on a purpose-built corpus (four self-validated non-inferable tasks + one inferable control, $\times$ three tiers $\times$ three reps; $n = 121$ verdicts; `analyze.mjs` recomputes every figure deterministically from the recorded `verdicts.tsv`). Each task is run under three spec states. (i) *Narrow* (the omitted edge stripped, no abstain option): non-inferable confident-false-pass = **94%** (opus 100%, sonnet 83%, haiku 100%; $n = 32$)—the $n = 12$ blind spot, reproduced and powered. (ii) *Surfaced-resolved* (the omitted edge written into the spec as an explicit criterion): catch = **94%**, residual false-pass 6% (opus/sonnet 100%, haiku

83%)—resolving the edge converts blind-spot into catch, mechanizing verifier-reach = spec-reach. (iii) *Surfaced-unresolved* (the *real* compiled edge-probe tag appended, abstention available): 33% false-pass, 47% abstain, 19% catch, tier-gated—opus 100% honest (abstain+catch), sonnet 67%, haiku 33% (flag-deaf). Two further results matter: the residual false-pass concentrates exactly on the one task the edge-probe’s recall *misses* (precision/tie: 0% catch even when surfaced—the $\sim 25\%$ recall gap where a held-out backstop is irreplaceable), and the inferable control shows 100% catch with 0% over-abstention—the real tagger, unlike a deliberately-false oracle, did not induce needless deferral.

Independent corroboration. The replication above lived in a personal fork the framework’s CI could not reach, so the maintainer independently rebuilt and committed an in-CI behavioral eval (`tests/verdict-eval/`): a 43-item corpus—vetted item-by-item by an independent model for blind-safety, realism, correct ground truth, and correct class (inferable 12, domain-knowledge 10, spec-silent 14, clean 7)—recorded blind on two models (Sonnet, Haiku) across three modes, 258 judgments, 0 malformed, 65 CI tests green. Recordings are committed transcripts replayed by a deterministic token-regex judge, so the result is key-free and reproducible. Two harnesses—ours and the maintainer’s, blind to each other, different models and prompts—land the same conclusion (Table 3).

This run also sharpens the construct: “non-inferable” is not one thing. It splits into a *domain-knowledge* class (a boundary a capable self-check can surface from its own training, e.g. banker’s rounding) and a *truly spec-silent* class (genuinely ambiguous without the spec). The blind spot proper is the spec-silent class, and only exogenous tagging closes it: spec-silent catch rises from 36% (Sonnet) / 43% (Haiku) at baseline to 79% / 71% under exogenous tags, and the confident-false-pass rate drops from 64% / 57% to 21% / 29%. Endogenous *self-disconfirmation* (re-derive criteria, then rebut your own verdict) is inert-to-harmful on the capable tier: net zero on Sonnet, where it *regressed* a correct flag (`pagination-index-base`, FLAG→PASS—the self-check talked itself out of a real catch); on Haiku it gives a modest +7–10pp. A distinct endogenous *abstention* mode (“don’t pass unless confident”; a fourth #1637 arm, not in Table 3) is a cheap routing win—spec-silent catch 36% → 57% (Sonnet) and 43% → 64% (Haiku), +21pp on both, at zero over-abstention cost—but it routes rather than resolves: it is the verify-time trigger for exogenous resolution, not a substitute for it.

Table 3: Powered, CI-committed eval (maintainer-built; 43 items, 2 models, 258 blind judgments). Recall = defect-catch rate per class; CFP = confident-false-pass on the spec-silent class. figures from open-gsd/gsd-core#1637 (`tests/verdict-eval/`).

model	mode	inferable	domain-know.	spec-silent	spec-silent CFP	clean prec.
Sonnet	baseline	100%	80%	36%	64%	100%
Sonnet	endogenous (disconf.)	100%	80%	36%	64%	100%
Sonnet	exogenous (tag)	100%	100%	79%	21%	100%
Haiku	baseline	100%	60%	43%	57%	86%
Haiku	endogenous (disconf.)	100%	70%	50%	50%	86%
Haiku	exogenous (tag)	100%	100%	71%	29%	100%

Hygiene. Two contamination corrections were applied mid-experiment and recorded: a function-name leak and a false-positive confound on the rounding task (segregated), plus comment stripping

of ground-truth hints. A separate cross-version run showed that leaving answer hints in the corpus inflated the apparent catch rate roughly $3\times$ (from 3/9 to 1/9 once sanitized)—a reminder that the honest number is the lower one.

3 Verifier Reach Is Spec Reach

Why is a non-inferable defect *unverifiable* rather than merely hard? Because the verifier has nothing to check against. Goal-backward verification compares the implementation to the specification’s stated obligations plus general convention. When the obligation was never stated and no convention determines it, the comparison is vacuous: any behavior is “consistent with the spec.” The verifier is not failing to reason; it is reasoning correctly over an input that does not contain the answer. This is why a held-out test and a strong goal-backward verifier “draw from the same source—the spec”: across the corpus, held-out tests add no marginal catch *on inferable defects*, and their only demonstrated marginal value is exactly on the non-inferable residue.

The implication inverts the usual remedy. Rather than strengthen the reader, surface the missing obligation into the spec. We observe the converse directly: when a weak overseer is handed structured `must_haves` that name the spec-defined correctness condition, it matches a strong reviewer on spec-defined defects—the artifact, not the model, supplies the reach. So the highest-value investment is spec completeness, with a held-out test as the backstop for edges the author knows but cannot fully articulate.

The general law. The non-inferable blind spot is the special case of a pipeline-wide pattern: *each stage’s reach equals its input artifact’s reach*. A planner cannot integrate a seam the plan never pinned (in a controlled run, independent executor waves integrated 0/9 on an unpinned interface versus 9/9 when the plan pinned it); a verifier cannot catch what the spec omitted; a gray-area resolver cannot resolve what the criteria never named. The failure mode at every stage is what the artifact silently got wrong or left out—never what it left visibly blank—and the fix is always to widen or reconcile the artifact upstream, never to deploy a stronger reader downstream.

Connections to engineering folklore. Two well-worn observations sharpen the framing. Goodhart’s law—when a measure becomes a target, it ceases to be a good measure—predicts the failure precisely: the moment an agent loop optimizes for “the verifier returns pass,” the verdict stops tracking correctness, and it stops first exactly where the verifier is blind. Our defenses are the standard Goodhart defenses—pair the proxy (the verdict) with an outcome measure that is harder to fake (a held-out test), and audit the proxy-to-reality relationship (the exogenous non-inferable tag). Linus’s law—“given enough eyeballs, all bugs are shallow”—fails for this bug class for the same reason it usually succeeds: it relies on cognitive diversity, but a non-inferable edge is an *assumption every reviewer shares*, because the missing information is absent from the artifact rather than hidden in the code. More model-eyeballs (an ensemble, a debate) cannot make shallow a bug that is invisible to all of them; the review that helped in practice (§5) was a human bringing outside knowledge the artifact lacked.

4 System

The pipeline (Figure 1) has two spec-time instruments that write a single verification-tiered contract, and one verify-time disposition that fails closed.



Figure 1: Pipeline: two spec-time probes write a verification-tiered **must_haves** contract; the verifier hard-gates mechanically-checkable items and abstains on what it cannot confirm.

4.1 Edge-probe: shape-classified completeness

At spec time, the edge-probe classifies each requirement by *shape* (numeric-range, collection, text, stateful, I/O) and, from a fixed eight-category checklist (boundaries, adjacency/touching, empty/degenerate inputs, encoding, ordering/stability, precision/overflow, idempotency, concurrency), raises the edges that the requirement’s shape implies. Each raised edge gets a disposition: *specify* (becomes an acceptance criterion), *backstop* (a held-out test stands in; the edge is tagged non-inferable), *dismiss* (with a recorded reason), or *defer* (an explicit planner assumption). The category taxonomy is sound; the binding weakness is the prose→shape classifier (§7). The instrument is most distinct from sampling-based clarification (§6) in that it emits a *persistent, verifiable contract* rather than a one-shot question.

4.2 Prohibition-probe: adversarial must-NOT elicitation

Shape classification surfaces what a requirement should *handle*; it is silent on what a feature must *not become*. A second instrument elicits prohibitions in two stages. Stage 1 (recall) asks adversarially, per requirement: “*what could this feature silently become that the author would not want, but the spec does not forbid?*”—over-generating candidates. Stage 2 (precision) runs a one-pass classifier that keeps only the genuine values/safety prohibitions and drops routine engineering must-nots. Measured: the adversarial framing reaches 25/25 holistic prohibitions and 17/17 canon-less (product-values) prohibitions *including* haiku at 6/6, where a role-played “three amigos” review reaches only 9/17 on canon-less cases; the classifier collapses a ~ 9.3 -item list to ~ 2.3 while retaining the ground-truth prohibition 8/8 and promoting 0/8 pure-engineering items to prohibitions on a clean battery. Confirmed prohibitions become **must_haves.prohibitions** entries.

Crucially, the difficulty of prohibitions is *not* a negation problem. When the same violating code is shown to verifiers under positive (“must”) versus negative (“must NOT”) framings, catch rate is identical ($9/9 = 9/9$, gap 0.00, ≈ 0.99 confidence). The real difficulty is **scope**: a prohibition quantifies universally over the repository, while a per-phase inspector samples. The remedy is therefore a repository-wide deterministic check, not a more negation-aware reader.

4.3 Fail-closed verification and abstention

Every contract item carries an orthogonal **verification tier**. *Test-tier* items (mechanically checkable by a negative test, lint, or AST rule) **hard-gate** the verify phase and fail closed if the check cannot be run. *Judgment-tier* items (irreducible values rules) and *backstop*-tagged truths route, when unconfirmable, to a distinct verdict **insufficient_spec** → **human_needed**—the verifier abstains rather than emitting a confident green. The disposition ships in the probe core as

the truth-axis counterpart to the verified prohibition disposition `dispositionForProhibition()` (`probe-core.cts`).

The decisive finding is that abstention must be **exogenous**. Table 4 shows the confident-false-pass rate on the blind spot by condition: a tag that *forces* deferral drops it from 100% to 17%, while an “abstain if you are unsure” instruction (endogenous) reaches only 67% and fires only on the *salient* ambiguity it happens to notice—never on the true blind spot, which the model cannot feel. Two costs bound the win: a *false* non-inferable tag makes a strong model over-abstain on a real spec-determined bug (33% over-abstention), so detector precision is load-bearing; and the weakest tier (haiku) is flag-deaf, heeding the tag inconsistently. Abstention is therefore exogenous *and* tier-pinned to a capable model. It also does not *recalibrate*: ECE among the verdicts that stay decisive on non-inferable tasks nudges from 0.93 to 0.98. The deliverable is fewer decisions in the blind spot, not better-calibrated ones. These are direction-finding numbers ($n = 27$). At scale the powered replication (§2, $n = 121$) shows the same shape with a *real* edge-probe tag: surfaced-unresolved drives 47% honest abstention, tier-gated (opus 100% honest, haiku 0%), and surfaced-*resolved* reaches 94% catch; the independent CI-committed eval (Table 3) corroborates—exogenous tagging lifts spec-silent catch to 71–79% and roughly thirds the confident-false-pass rate—and sharpens one caveat: endogenous *self-disconfirmation* (as opposed to abstention) is not merely weak but can be *harmful*, regressing a correct flag on the capable model. Abstention routes the case to exogenous resolution; it does not diagnose, and self-disconfirmation should not ship as a verification control.

Table 4: Confident false-pass on the non-inferable blind spot, by abstention condition ($n = 27$, 1 rep).

Condition	n	false-pass	abstain (honest)	mean conf. of false-pass
Baseline (no abstain option)	6	100%	0%	0.93
Endogenous (self-judged abstain)	6	67%	33%	0.96
Exogenous (tagged non-inferable)	6	17%	83%	0.98

A useful property of exogenous abstention: it routes correctly *even with the wrong diagnosis*. Under tagging, verifiers abstained on the grapheme task while citing unrelated reasons (a negative `max`, a null coercion)—they identified the wrong gap and still took the right action. A mechanism that produces correct *deferral* without correct *identification* is exactly what a blind spot needs, because if the verifier could name the true edge it would not be blind.

4.4 The unifying primitive

Edges and prohibitions look like different features, but they share one structure: each contract item carries a `verification` tier, and *abstention is the disposition when that tier cannot be discharged*. Mechanically checkable obligations hard-gate; unconfirmable ones abstain. This collapses “surface the gap,” “write the must-NOT,” and “don’t fake the green” into a single contract with a single fail-closed rule.

5 Deployment

The pipeline ships in the open-source GSD-Core framework. As of this draft the merged pieces are: the edge-probe (issue #550 / PR #584); recall sharpening (#1108) and unclassified-edge surfacing (#1117); the prohibition-probe (#644 / PR #1149); test-tier enforcement, the fail-closed

hard-gate (#1273); the deterministic check-descriptor that lets the verifier locate a check rather than invent one (#1301); and the honest verifier / truth-axis abstention (#1154 / PR #1738). A machine-proven fail-first guard (#1314) is in flight.

We treat the maintainer review itself as an external validation signal. Across many rounds, an independent maintainer hit both failure modes the program had predicted before adopting the fix: spec prose-drift, and a self-graded review that rationalized away its own findings (observed as malicious compliance on a separate PR). The test-tier enforcement that now hard-gates mechanically-checkable prohibitions was seeded directly by a maintainer’s observation that the prohibition-probe’s safety half shipped before its enforcement half. This is one ecosystem and one maintainer—weak evidence in isolation—but it is adversarial, independent, and it moved the design.

The validation went further than review. Because our headline abstention numbers lived in a personal fork and were not reproducible from the framework’s repository, the maintainer independently built the powered, CI-committed eval of §2 (Table 3) to re-test the claim from scratch—a separate corpus, separate models, blind to ours. It corroborated the load-bearing result (exogenous tagging closes the spec-silent gap), refined the construct (the domain-knowledge vs spec-silent split), and *overturned* an encouraging first-run signal (endogenous self-disconfirmation, 0% → 67% at $n = 3$, collapsed to inert-or-harmful when powered). An independent party re-deriving, sharpening, and partially falsifying a claim is the strongest external signal in this paper.

6 Related Work

Spec completeness and clarification. ClarifyGPT [6] detects ambiguity by sampling code completions and asking clarifying questions where they disagree. The non-inferable blind spot is precisely where samples *agree* on the wrong behavior, so consistency-sampling is silent there; the edge-probe is shape-driven and emits a persistent, verifiable contract rather than a one-shot question. Boundary Value Analysis [7] is a decades-old test-design technique; our placement of it *upstream, at spec-authoring time, as a completeness gate wired into the verifier* is the repositioning, not the technique.

Negative and prohibition constraints. Omission-constraint decay [2] shows prohibitions are fragile under context (adherence 73% → 33% by turn 16 over thousands of trials), motivating a persistent contract over an in-prompt instruction. Enforcement frameworks such as Safety Chip [8] compile constraints into checkable form. Our distinction is spec-time elicitation paired with tiered, fail-closed verification.

Abstention and selective prediction. Conformal Abstention [1] offers hallucination-rate guarantees via conformal prediction. AbstentionBench [4] benchmarks abstention on *underspecified* inputs and finds it unsolved, with “scaling models of little use”—which independently corroborates our model-invariant blind spot. Our distinction is a coding-agent verify-phase abstention gated by an *exogenous* inferability tag (not self-judged confidence) and emitted as a first-class `insufficient_spec` verdict.

LLM-as-judge weakness. Studies of verification for code generation report that homogeneous or limited test suites let subtle faults go undetected and recommend held-out tests [5]; surveys catalog LLM-as-judge weaknesses [3]. These corroborate verifier overconfidence; the spec-reach

framing—and the specific claim that overconfidence *concentrates* on non-inferable requirements—is ours.

7 Limitations

- **Sample sizes; still not a large study.** Our calibration result is $n = 12$; the powered replication is $n = 121$ but over only four non-inferable tasks (so the $\sim 25\%$ recall-gap estimate rests on 1/4 categories) and three reps; the independent corroboration is $n \approx 10$ –14 per class, two models, one run/seed. The spec-silent class has no ground-truth “right answer” without the spec, so even exogenous tagging cannot reach 100% there, and the inferable/non-inferable label is itself model-dependent. Effects are large and monotone, but we report leads plus reproducible harnesses, not a powered multi-seed study, and there is no preregistration.
- **Recall residual.** The shape-classification step is a leaky abstraction: on prose alone the deterministic classifier under-fires ($\sim 48\%$ of applicable edges missed on one corpus, worse on a terser second corpus), because terse prose yields no shape and therefore no edge. An LLM-in-the-loop classifier recovers most of this on a sonnet-class model or better (under-fire $\sim 14\%$ at sonnet, near-zero at opus), but the weakest tier is high-variance and unreliable. On an edge the probe fails to *name*, the verifier still catches 0%—so the pipeline is not a complete fix, and a held-out backstop remains necessary for the residue.
- **Abstention is not free.** A false non-inferable tag induces over-abstention on real spec-determined bugs; the weakest tier is flag-deaf. Abstention quality is bounded by detector precision and model tier.
- **Single ecosystem.** All deployment evidence is from one framework and one maintainer. Any cross-model or cross-family verifier recommendation is *unvalidated*—every verifier subagent in these experiments was from a single model family.
- **Model-version sensitivity.** Abstention is tier-gated and confidence-while-wrong varies by tier; results may shift across model generations.

8 Conclusion

The overconfident verifier is not a reasoning failure to be out-reasoned; it is a reach failure to be out-specified. A verifier catches what the spec named, so the productive intervention is to name more, earlier—shape-classified edges and adversarially-elicited prohibitions written into a verification-tiered contract—and to make verification fail honestly, abstaining on what it cannot confirm rather than stamping a confident green. We have shipped this into an open-source framework and measured where it works and where it does not. The honest bottom line is a synthesis with an original measurement and a residual that still needs a held-out backstop—move defect-naming forward, make verification fail closed, and keep the oracle for the edges no spec yet reaches.

Reproducibility. The corpora, harnesses, and the merged feature PRs are public. The $n = 121$ replication ships its recorded `verdicts.tsv` and `analyze.mjs`, which recomputes every figure deterministically (no model calls). The independent corroboration is a committed, key-free, CI-green artifact in the framework repository (`tests/verdict-eval/`, `open-gsd/gsd-core#1637`): committed transcripts replayed by a deterministic judge, per-slice metrics locked as a golden fixture—anyone

can re-run it. Every figure in this paper traces to a committed primary that recomputes deterministically; the smallest, single-rep results would still benefit from additional reps and confidence intervals, a camera-ready refinement rather than an open question of support.

Acknowledgements

The research thesis and measurements are the author’s own work. The pipeline features ship in the open-source GSD-Core project and are community contributions; merges and review were performed by the project’s maintainers, whose adversarial review materially improved the design.

References

- [1] Yasin Abbasi-Yadkori, Ilja Kuzborskij, David Stutz, András György, Adam Fisch, Arnaud Doucet, Iuliya Beloshapka, Wei-Hung Weng, Yao-Yuan Yang, Csaba Szepesvári, Ali Taylan Cemgil, and Nenad Tomasev. Mitigating LLM hallucinations via conformal abstention, 2024. Conformal-prediction abstention with theoretical guarantees on the hallucination rate.
- [2] Yeran Gamage. Omission constraints decay while commission constraints persist in long-context LLM agents, 2026. Omission compliance 73% (turn 5) $\rightarrow$ 33% (turn 16) while commission holds at 100% (Mistral Large 3, $p < 10^{-33}$); 4,416 trials, 12 models, 8 providers.
- [3] Junda He, Jieke Shi, Terry Yue Zhuo, Christoph Treude, Jiamou Sun, Zhenchang Xing, Xiaoning Du, and David Lo. LLM-as-a-judge for software engineering: Literature review, vision, and the road ahead, 2025. Literature review of LLM-as-judge for evaluating software-engineering artifacts.
- [4] Polina Kirichenko, Mark Ibrahim, Kamalika Chaudhuri, and Samuel J. Bell. AbstentionBench: Reasoning LLMs fail on unanswerable questions, 2025. Evaluating 20 frontier LLMs, “abstention is an unsolved problem, and one where scaling models is of little use.”.
- [5] Zihan Ma, Taolin Zhang, Maosong Cao, Junnan Liu, Wenwei Zhang, Minnan Luo, Songyang Zhang, and Kai Chen. Rethinking verification for LLM code generation: From generation to testing, 2025. Limited/homogeneous test suites let subtle faults go undetected and inflate measured performance; proposes SAGA for more comprehensive test generation.
- [6] Fangwen Mu, Lin Shi, Song Wang, Zhuohao Yu, Binqun Zhang, Chenxue Wang, Shichao Liu, and Qing Wang. ClarifyGPT: Empowering LLM-based code generation with intention clarification. In *Proceedings of the ACM on Software Engineering (FSE)*, 2024. arXiv:2310.10996. Code-consistency ambiguity detection + clarifying questions; GPT-4 70.96% $\rightarrow$ 80.80% on MBPP-sanitized.
- [7] Glenford J. Myers, Corey Sandler, and Tom Badgett. *The Art of Software Testing*. Wiley, 3rd edition, 2011. Boundary Value Analysis as a classical test-design technique.
- [8] Ziyi Yang, Shreyas S. Raman, Ankit Shah, and Stefanie Tellex. Plug in the safety chip: Enforcing constraints for LLM-driven robot agents, 2023. A queryable linear-temporal-logic safety module: NL $\rightarrow$ LTL encoding, violation reasoning, and unsafe-action pruning for LLM-driven agents.